

Tutorial series 1

Exercise 1:

1. Amina wants to write a program that uses the frequency of the words "science," "public," "access," "university," "government," "funding," "education," "budget," "justice," and "law" to determine whether an article deals with science policy or not. She started by annotating a thousand articles according to their subject. What kind of machine learning problem does she need to solve?
2. Among the following problems, which ones are well-suited to be addressed by machine learning?
 - Determining the optimal time to post content on a webpage
 - Determining the shortest path between two nodes in a graph
 - Predicting the number of bikes to allocate to each station in a city bike-sharing system
 - Estimating the price a masterpiece painting could reach at an auction
 - Denoising a signal
3. Given a program that uses banking data to detect bank fraud, what are E, T, and P?
4. What are the advantages and disadvantages of a simple model compared to a complex model?
5. A person wants to automatically identify which of their vacation photos were taken indoors, outdoors, during the day, or at night. Should they use 4 binary classifiers, 2 binary classifiers, or 1 binary classifier?
6. Given a bank fraud detection algorithm, an undetected fraud costs the bank more than manually processing a non-fraudulent transaction. Should we prioritize the false positive rate, false negative rate, or true positive rate?
7. Given a binary classification algorithm that labels every observation as 'positive,' the evaluation dataset contains 85% positive examples and 15% negative examples. What will be the accuracy, recall, and precision?
8. A binary classification algorithm randomly returns negative or positive with a probability of 0.5 for each class. The evaluation dataset contains 85% positive examples and 15% negative examples. What will be the accuracy, recall, and precision?
9. Bilal trained a binary classification model on 1000 observations. His model makes mistakes on 10 of these observations. Is he in a situation of underfitting, overfitting, or neither?

Exercise 2

Consider the linear regression model of the form:

$$f(x) = \beta_0 + \sum_{j=1}^p \beta_j x_j$$

The coefficients are obtained by minimizing the sum of least squares, namely:

$$\arg \min \sum_{j=1}^n \left(y^i - \beta_0 - \sum_{j=1}^p \beta_j x_j \right)^2$$

By transforming the problem into matrix form by adding a column of 1s to the matrix X, and assuming that the matrix X is invertible (rank = p+1), the sum of least squares is minimized for:

$$\beta^\square = (X^t X)^{-1} X^t y$$

Now consider 5 observations in two dimensions, labeled by y:

X1	0.12	0.14	0.31	0.37	0.49
X2	29.0	33.0	17.0	21.0	12.0
y	21.0	24.3	12.7	15.6	9.0

1. What are the parameters of a linear regression in this case?
2. What is the mean squared error of the model on this data?
3. What happens if we multiply x1 by 10 and subtract 1 from it?

Exercise 3

In a clinical study, 27 patients are observed, and their weight, age, gender (male/female), and the presence of surgery in their medical history (yes/no) are recorded. Then, their score on a physical test is measured (continuously scored between 0 and 20, with 20 representing maximum performance). The goal of the learning task is to predict the score based on weight, age, gender, and the presence of surgery.

1. Formalize this study as a learning problem within the course framework. Specify d, n, the nature of the variables (quantitative or qualitative), and whether it is a regression or classification problem. Also, explicitly define the input space X.

Consider two patients. The first is a 40-year-old woman weighing 60 kg with no history of surgery. The second is a 30-year-old man weighing 80 kg with no history of surgery. Calculate the distance D between these two patients. D